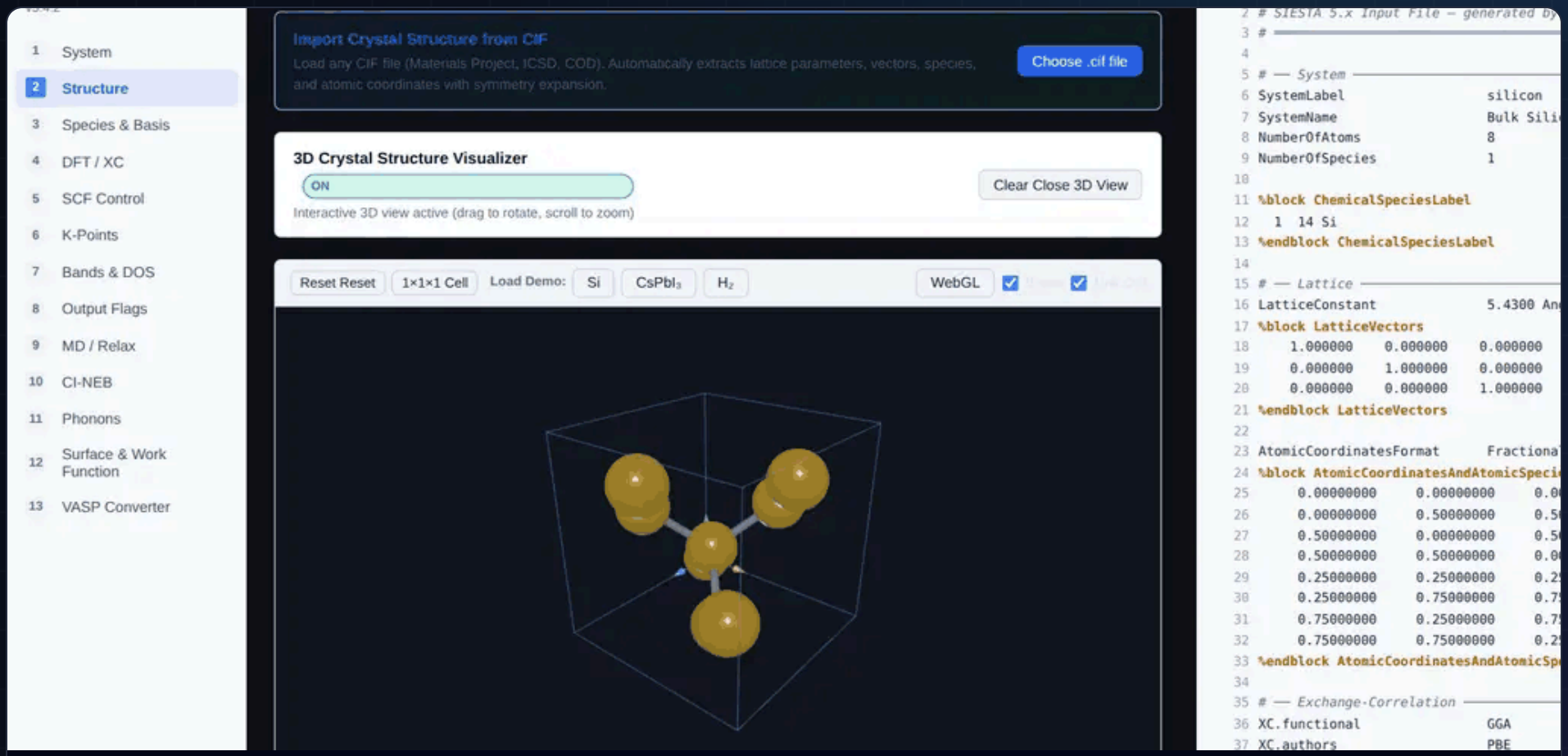
SIESTA Input Builder & Post-DFT Studio

Zero-install interactive GUI: from raw CIF/POSCAR to production FDF in seconds with complete client-side privacy.



1-Click CIF Import
Extracts parameters, vectors, elements, and fractional symmetry positions.

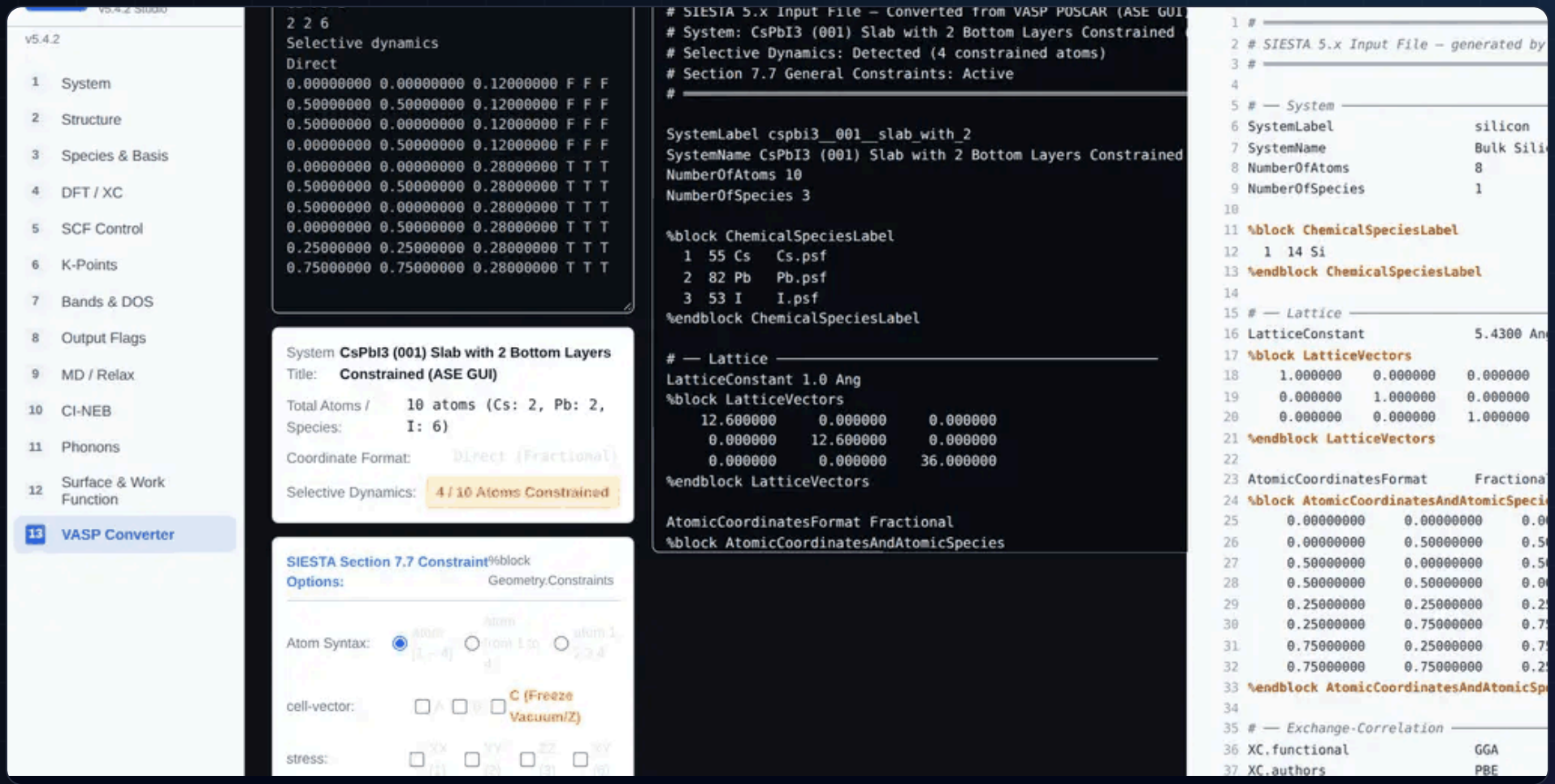
3D WebGL Crystal Viewer
Rotate, pan, zoom with real-time magnification scale and high-res PNG snapshot.

Open Source • 100% In-Browser Privacy

`chemsiva.github.io/siesta-input-builder/`

VASP POSCAR to SIESTA FDF Converter

Ingest structures with ASE GUI Selective Dynamics and auto-generate Section 7.7 Geometry.Constraints.



Selective Dynamics Parser
Translates F F F into atom ranges and T T F into directional Z-only constraints.

1-Click Workspace Transfer
Loads the converted structure straight into Section 2 and the 3D crystal viewer.

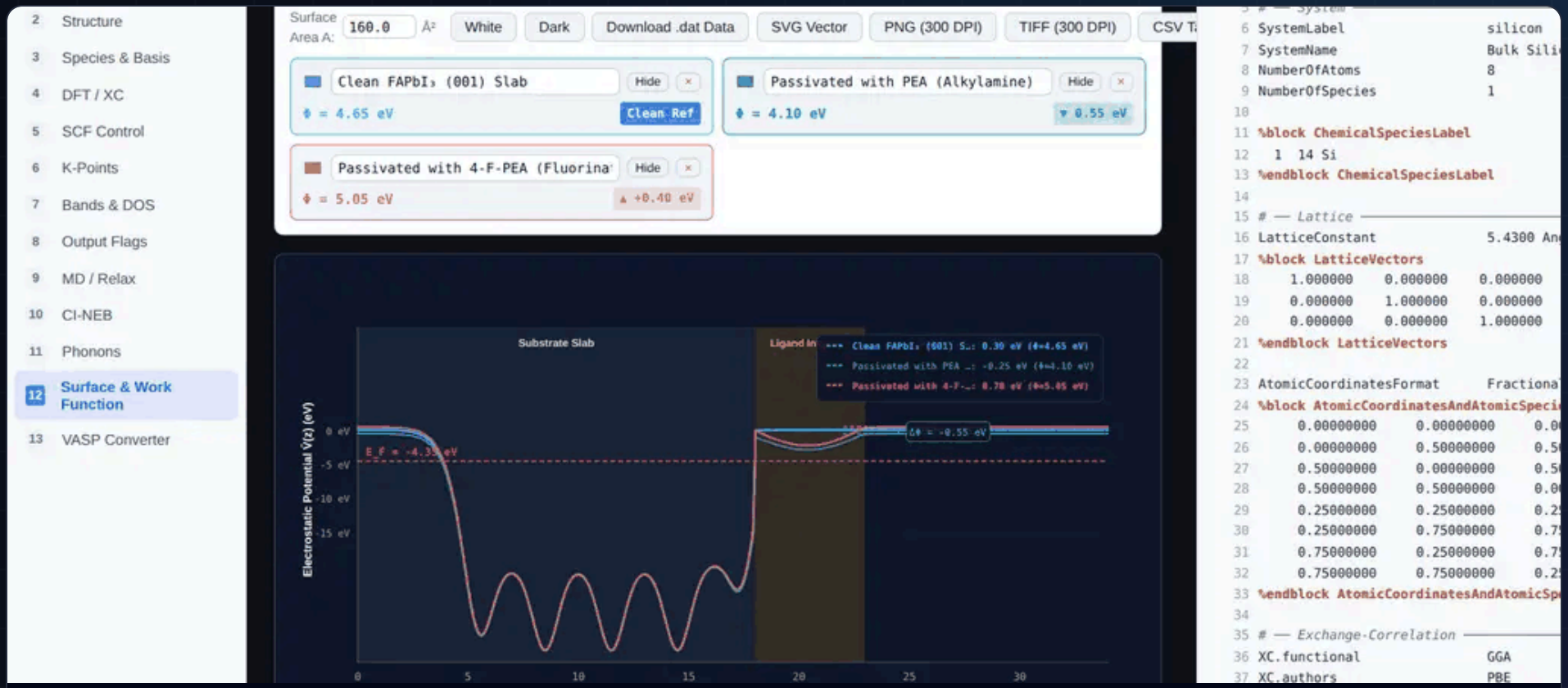
Section 7.7 General Constraints Supported

`chemsiva.github.io/siesta-input-builder/`

SURFACE SCIENCE | Work Function Studio

Electrostatic Potential & Work Functions

Extract asymptotic vacuum plateaus, quantify work function modification, and calculate interface dipoles.



Multi-Surface Comparison
Overlay clean vs passivated slabs with automated Delta-Phi work function steps.

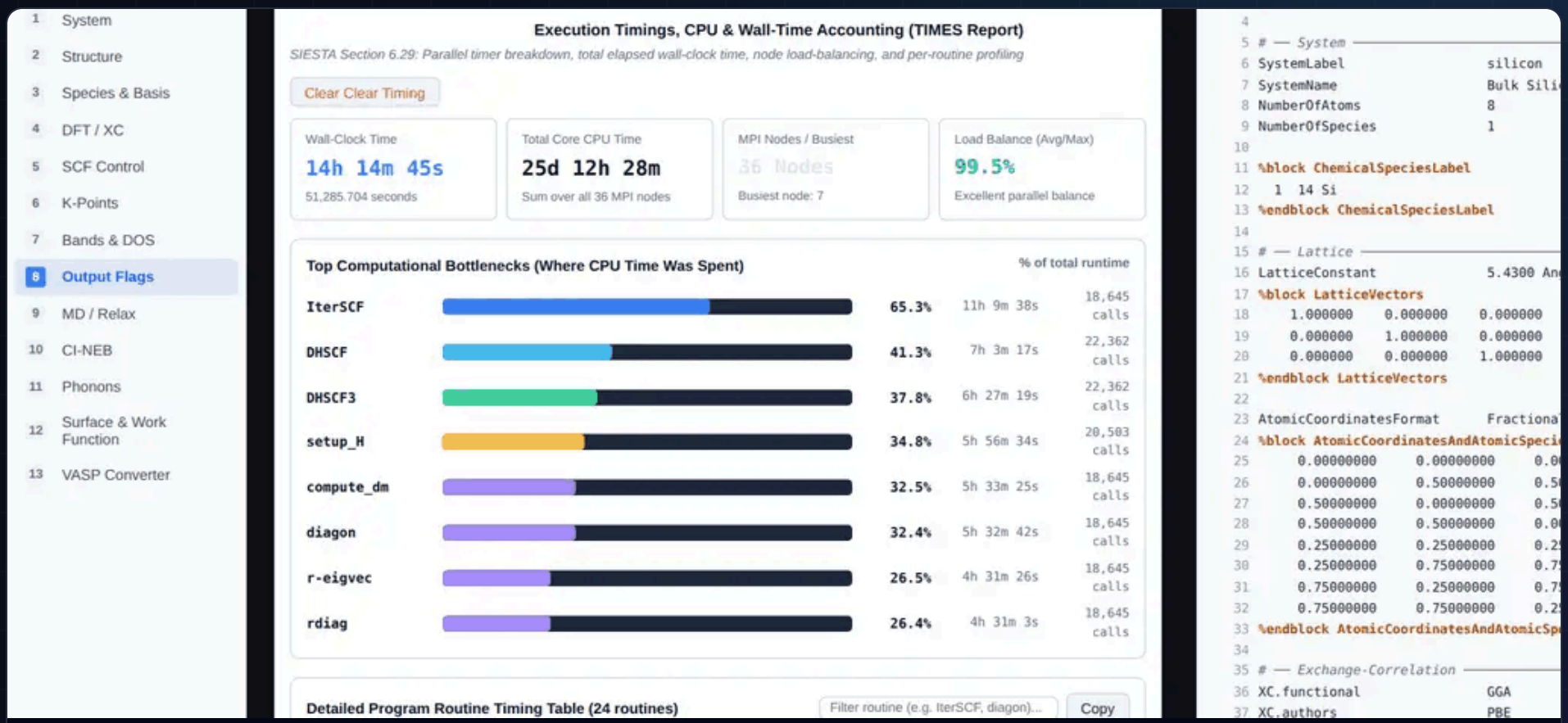
3D Band Alignment Diagrams
Construct publication-grade energy band diagrams for perovskites and solar cells.

Export Vector SVG, 300 DPI PNG, 300 DPI TIFF

chemsiva.github.io/siesta-input-builder/

HPC TIMES Profiler & Reaction Suites

Benchmark parallel scalability, generate CI-NEB reaction packages, and build frozen-phonon matrices.



Section 6.29 TIMES Analyzer
Inspect node load balancing, total core CPU hours, and routine bottlenecks.

CI-NEB & Phonons
Automate transition state searches, IDPP interpolation, and Vibra dispersion runs.

Free & Open Source • 19-Page PDF Guide Included

`chemsiva.github.io/siesta-input-builder/`